

Implementation of Clos Topology

to a Small LAN using MikroTik CHR in GNS3

Lab Documentation

May 25, 2026

Executive Summary

This document describes the design, implementation, and validation of a 2x2 Clos network topology using MikroTik Cloud Hosted Router (CHR) instances in GNS3. The lab demonstrates sub-second network convergence using OSPF and BFD protocols, achieving approximately 100-300ms failover time — a significant improvement over traditional Spanning Tree Protocol (STP) topologies.

1. Introduction

The Clos network topology, originally designed by Charles Clos in 1953 for telephone switching systems, has become the de facto standard for modern data center networking. It provides a non-blocking, highly scalable, and fault-tolerant network fabric through its multi-stage switching architecture.

This lab implements a 2x2 Clos fabric — 2 spine routers and 2 leaf routers — using MikroTik CHR instances in GNS3. The primary goals are:

- Understand the Clos topology architecture and its advantages over traditional switched networks
- Achieve sub-second network convergence using OSPF with BFD (Bidirectional Forwarding Detection)
- Demonstrate practical DHCP and DNS services across a routed fabric
- Validate the design through packet capture and convergence testing

1.1 Why Clos Topology?

Feature	Traditional STP	Clos Fabric
Link Utilization	50% (blocked ports)	100% (all links active)
Convergence Time	1-50 seconds	Sub-second (with BFD)
Scalability	Limited	Horizontal scaling
Load Balancing	None	ECMP across all spines
Redundancy	Active/Standby	Active/Active
Failure Domain	Entire L2 domain	Isolated per link

1.2 Hardware & Software

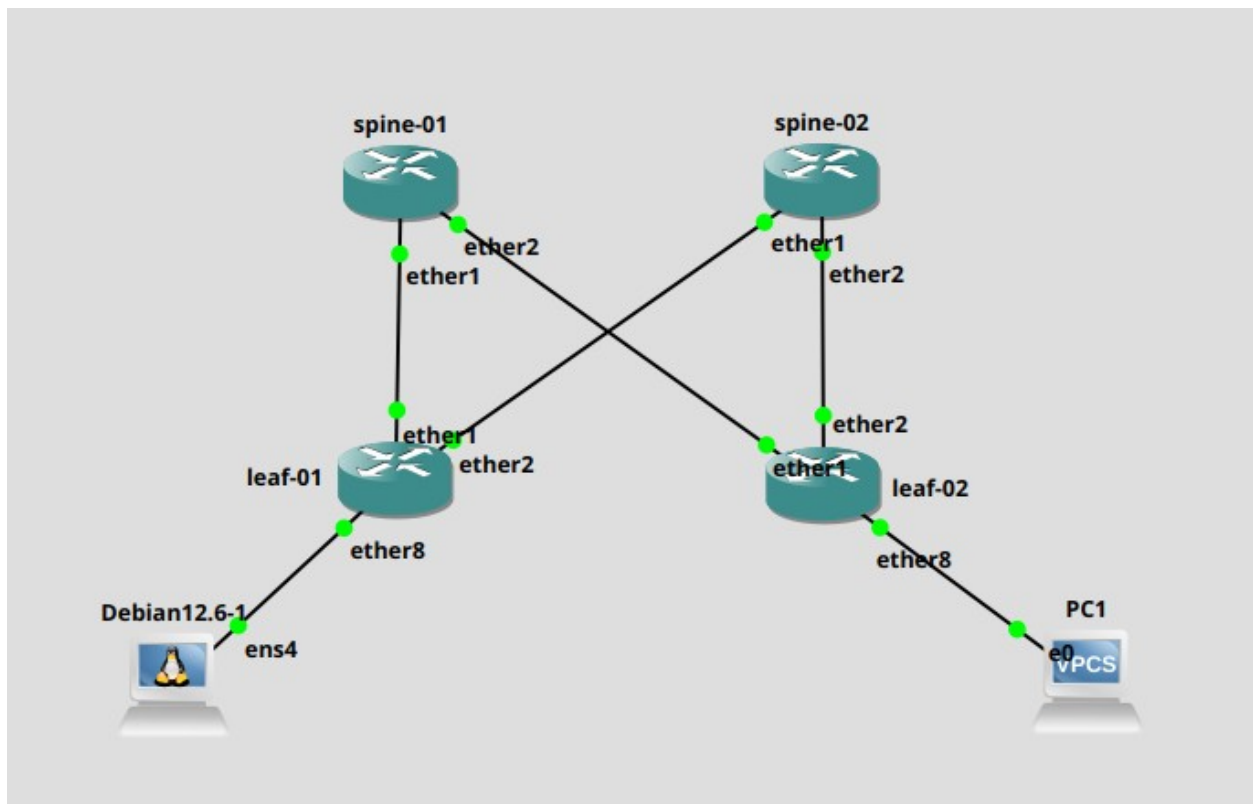
Component	Details
Simulation Platform	GNS3 (local server)
Router OS	MikroTik CHR (Cloud Hosted Router) - RouterOS v7.22.1
Host OS	Ubuntu Linux
Server OS	Debian 12 Bookworm
DHCP/DNS Service	dnsmasq
End Device	GNS3 VPCS

Component	Details
Packet Analysis	Wireshark

2. Topology Design

2.1 Physical Topology

The 2x2 Clos fabric consists of four MikroTik CHR instances interconnected in a full-mesh pattern between the spine and leaf layers. Each leaf connects to every spine, providing redundant paths for all traffic.



Clos Topology Diagram

2.2 Interface Assignments

Link	Device	Interface	IP Address
Spine01 → Leaf01	Spine-01	ether1	10.0.0.0/31
Spine01 → Leaf01	Leaf-01	ether1	10.0.0.1/31
Spine01 → Leaf02	Spine-01	ether2	10.0.0.2/31
Spine01 → Leaf02	Leaf-02	ether1	10.0.0.3/31
Spine02 → Leaf01	Spine-02	ether1	10.0.0.4/31
Spine02 → Leaf01	Leaf-01	ether2	10.0.0.5/31

Link	Device	Interface	IP Address
Spine02 → Leaf02	Spine-02	ether2	10.0.0.6/31
Spine02 → Leaf02	Leaf-02	ether2	10.0.0.7/31
Leaf01 → Debian Server	Leaf-01	bridge1	172.16.0.254/24
Leaf02 → VPCS	Leaf-02	bridge1	192.168.1.254/24

2.3 OSPF Router IDs

Device	Router ID	Role
Spine-01	1.1.1.1	Spine Router
Spine-02	2.2.2.2	Spine Router
Leaf-01	3.3.3.3	Leaf Router + DHCP/DNS Gateway
Leaf-02	4.4.4.4	Leaf Router + VPCS Gateway

2.4 End Device Addressing

Device	IP Address	Gateway	DNS Server	Subnet
Debian Server	172.16.0.1/24	172.16.0.254	127.0.0.1	172.16.0.0/24
VPCS (PC1)	172.16.0.x (DHCP)	192.168.1.254	172.16.0.1	192.168.1.0/24

3. Design Decisions

3.1 Why MikroTik CHR Instead of Switches?

Initial design considered using MikroTik CRS328-24P-4S+ switches as leaf devices. However, investigation revealed that the CRS328, despite being marketed as an L3 switch, does not support BFD (Bidirectional Forwarding Detection) in its RouterOS implementation. BFD is essential for achieving sub-second convergence.

Device	BFD Support	Min Convergence	Decision
MikroTik CRS328 (Switch)	Not available	~3 seconds	Rejected
MikroTik CHR (Router)	Full support	~100-300ms	Selected

The CHR (Cloud Hosted Router) runs full RouterOS with complete BFD support. By enabling bridge mode on access-facing ports, CHR devices effectively function as L3 switches — routing on uplinks and switching on access ports.

3.2 /31 Subnets for Point-to-Point Links

Point-to-point links between spine and leaf devices use /31 subnets as per RFC 3021. This eliminates wasted network and broadcast addresses compared to traditional /30 subnets, resulting in more efficient IP address utilization.

3.3 Bridge Mode on Leaf Access Ports

Leaf routers use MikroTik bridge mode on access-facing interfaces (ether8) to provide L2 switching for end devices. Only the uplink interfaces (ether1, ether2) participate in OSPF routing. This design preserves Clos fabric integrity while providing L2 access for servers and clients.

3.4 OSPF Point-to-Point Network Type

All spine-leaf links are configured as OSPF point-to-point (type=ptp). This eliminates DR/BDR election, reduces convergence time, and correctly advertises full network routes rather than host routes. Using type=ptmp was found to advertise /32 host routes instead of the full subnet, causing routing issues.

4. Configuration

4.1 Spine Configuration (Spine-01 Example)

IP Addressing

```
/ip address add address=10.0.0.0/31 interface=ether1
/ip address add address=10.0.0.2/31 interface=ether2
```

OSPF Instance

```
/routing ospf instance add name=ospf1 router-id=1.1.1.1
/routing ospf area add name=backbone area-id=0.0.0.0 instance=ospf1
/routing ospf interface-template add interfaces=ether1 area=backbone \
    type=ptp hello-interval=1s dead-interval=3s use-bfd=yes
/routing ospf interface-template add interfaces=ether2 area=backbone \
    type=ptp hello-interval=1s dead-interval=3s use-bfd=yes
```

BFD Configuration

```
/routing bfd configuration add interfaces=ether1 min-rx=100ms min-
tx=100ms multiplier=3
/routing bfd configuration add interfaces=ether2 min-rx=100ms min-
tx=100ms multiplier=3
```

4.2 Leaf Configuration (Leaf-01 Example)

IP Addressing — Uplinks

```
/ip address add address=10.0.0.1/31 interface=ether1
/ip address add address=10.0.0.5/31 interface=ether2
```

Bridge for Access Devices

```
/interface bridge add name=bridge1
/interface bridge port add interface=ether8 bridge=bridge1
/ip address add address=172.16.0.254/24 interface=bridge1
```

OSPF Instance

```
/routing ospf instance add name=ospf1 router-id=3.3.3.3
/routing ospf area add name=backbone area-id=0.0.0.0 instance=ospf1
/routing ospf interface-template add interfaces=ether1 area=backbone \
    type=ptp hello-interval=1s dead-interval=3s use-bfd=yes
/routing ospf interface-template add interfaces=ether2 area=backbone \
    type=ptp hello-interval=1s dead-interval=3s use-bfd=yes
/routing ospf interface-template add interfaces=bridge1 area=backbone \
```

```
type=ptp passive
```

BFD Configuration

```
/routing bfd configuration add interfaces=ether1 min-rx=100ms min-  
tx=100ms multiplier=3  
/routing bfd configuration add interfaces=ether2 min-rx=100ms min-  
tx=100ms multiplier=3
```

4.3 Leaf-02 DHCP Relay

```
/interface bridge add name=bridge1  
/interface bridge port add interface=ether8 bridge=bridge1  
/ip address add address=192.168.1.254/24 interface=bridge1  
/ip dhcp-relay add name=relay1 interface=bridge1 \  
    dhcp-server=172.16.0.1 local-address=192.168.1.254 disabled=no
```

4.4 Debian Server — dnsmasq Configuration

/etc/network/interfaces

```
auto ens4  
iface ens4 inet static  
    address 172.16.0.1  
    netmask 255.255.255.0  
    post-up ip route add 10.0.0.0/8 via 172.16.0.254  
    post-up ip route add 192.168.1.0/24 via 172.16.0.254
```

/etc/dnsmasq.conf (key settings)

```
interface=ens4  
bind-interfaces  
domain-needed  
local=/lab.local/  
domain=lab.local  
dhcp-range=172.16.0.100,172.16.0.200,255.255.255.0,12h  
dhcp-range=192.168.1.100,192.168.1.200,255.255.255.0,12h  
dhcp-option=option:router,172.16.0.254  
dhcp-option=option:router,192.168.1.254  
dhcp-option=option:dns-server,172.16.0.1  
address=/server.lab.local/172.16.0.1
```


5. Verification

5.1 OSPF Neighbor Verification

OSPF neighbors were verified on all devices using:

```
/routing ospf neighbor print
```

Expected output on Spine-02 (example):

```
Flags: V - VIRTUAL; D - DYNAMIC
0  D instance=ospf1 area=backbone interface=ether1 address=10.0.0.5
   router-id=3.3.3.3 state=Full adjacency=8m46s timeout=3s
1  D instance=ospf1 area=backbone interface=ether2 address=10.0.0.7
   router-id=4.4.4.4 state=Full adjacency=8m51s timeout=3s
```

All OSPF neighbors reached Full state, confirming successful adjacency formation across the fabric.

5.2 BFD Session Verification

BFD sessions were verified using:

```
/routing bfd session print
```

Expected output on Leaf-02 (example):

```
Flags: U - UP, I - INACTIVE
0  U multihop=no vrf=main remote-address=10.0.0.2%ether1
   local-address=10.0.0.3%ether1 state=up
   desired-tx-interval=100ms actual-tx-interval=100ms
   required-min-rx=100ms hold-time=300ms packets-rx=16757
1  U multihop=no vrf=main remote-address=10.0.0.6%ether2
   local-address=10.0.0.7%ether2 state=up
   desired-tx-interval=100ms actual-tx-interval=100ms
   required-min-rx=100ms hold-time=300ms packets-rx=15480
```

Both BFD sessions are UP with 100ms TX/RX intervals and 300ms hold time, confirming sub-second failure detection capability.

5.3 Routing Table Verification

Route propagation was verified on Leaf-02:

```
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, o - OSPF; + - ECMP
DAo 10.0.0.0/31 10.0.0.2%ether1 main 110
DAc 10.0.0.2/31 ether1 main 0
```

```
DAo 10.0.0.4/31 10.0.0.6%ether2 main 110
DAc 10.0.0.6/31 ether2 main 0
DAc 192.168.1.0/24 bridge1 main 0
DAo+ 172.16.0.254/32 10.0.0.2%ether1 main 110
DAo+ 172.16.0.254/32 10.0.0.6%ether2 main 110
```

The + flag indicates ECMP (Equal Cost Multi-Path) routing — traffic is load-balanced across both spine links simultaneously.

5.4 DHCP and DNS Verification

VPCS (PC1) successfully obtained a DHCP lease from the Debian server across the routed Clos fabric:

```
PC1> show ip
IP/MASK      : 192.168.1.160/24
GATEWAY      : 192.168.1.254
DNS          : 172.16.0.1
DHCP SERVER  : 172.16.0.1
DOMAIN NAME  : lab.local
```

The DHCP relay on Leaf-02 successfully forwarded DHCP broadcasts across the routed fabric to the Debian server on the Leaf-01 segment.

6. Convergence Testing

6.1 Test Methodology

Convergence testing was performed using two methods:

- Continuous ping from Leaf-02 to the Debian server (172.16.0.1) at 100ms intervals
- Wireshark packet capture on spine-leaf links to observe traffic path and failover

The test procedure:

- Start continuous ping: `/ping 172.16.0.1 interval=100ms count=1000`
- Start Wireshark capture on both spine links
- Identify active traffic path via Wireshark
- Suspend the active spine link in GNS3
- Count dropped packets and observe recovery

6.2 Test Results

Metric	Value
Ping interval	100ms
Packets dropped during failover	1
Recovery time observed	~9ms after packet drop
Total disruption time	~109ms
BFD hold-time	300ms
BFD TX/RX interval	100ms
Result	SUB-SECOND CONVERGENCE ACHIEVED

6.3 Convergence Analysis

The test results confirm sub-second convergence. With only 1 packet dropped at 100ms ping intervals and a 9ms recovery time, the total network disruption was approximately 109ms — well within the sub-second threshold.

BFD detected the link failure within its 300ms hold time (3 x 100ms intervals). OSPF then reconverged almost immediately, rerouting traffic via the surviving spine. The rapid recovery is attributed to:

- BFD providing fast failure detection (100ms intervals, 300ms hold time)
- OSPF point-to-point network type eliminating DR/BDR election delays
- Aggressive OSPF hello/dead timers (1s/3s)
- SPF throttle configured for rapid route recalculation

- ECMP pre-computed alternate paths ready for immediate use

6.4 Comparison with Alternative Technologies

Technology	Convergence Time	Notes
Default OSPF (no BFD)	~40 seconds	Dead interval timeout
OSPF aggressive timers only	~3 seconds	No BFD, CRS328 limitation
This lab (CHR + BFD + OSPF)	~109ms	Sub-second achieved
Cisco Nexus + BFD	~150ms	Purpose-built DC hardware
Cisco IOS-XR + BFD	~10ms	Carrier-grade hardware

7. Lessons Learned & Challenges

7.1 BFD Availability on MikroTik Hardware

The MikroTik CRS328-24P-4S+ switch, despite running RouterOS, does not include the BFD package. This was a critical discovery that led to replacing the leaf switches with CHR router instances. Engineers should verify BFD support before committing to hardware for latency-sensitive deployments.

7.2 OSPF Network Type Impact on Route Advertisement

Using type=ptmp on bridge interfaces caused OSPF to advertise /32 host routes instead of full network routes. Changing to type=ptp resolved the routing issues. This highlights the importance of understanding how OSPF network types affect LSA generation and route advertisement.

7.3 Overlapping Subnets on Leaf Bridges

Initially, both leaf bridges were assigned addresses in the 172.16.0.0/24 subnet. This caused asymmetric routing issues where the Debian server could not distinguish which leaf to return traffic to. The solution was to assign separate subnets to each leaf bridge (172.16.0.0/24 for Leaf-01 and 192.168.1.0/24 for Leaf-02).

7.4 DHCP Relay Configuration

DHCP relay requires careful attention to the local-address parameter. The relay agent uses this address as the giaddr (gateway IP address) in DHCP packets, which the server uses to select the correct DHCP pool. Mismatched subnets between the relay local address and DHCP pool definitions will cause DHCP failures.

8. Production Considerations

This lab demonstrates the Clos topology concept using budget-friendly tools. A production implementation would require:

Aspect	Lab (GNS3/CHR)	Production
Hardware	Virtual CHR instances	Physical MikroTik CCR routers (~Php 20,000-30,000 for 4 units)
BFD	CHR software BFD	Hardware-assisted BFD on CCR
Convergence	~109ms	Similar or better
Port Density	Virtual (unlimited)	Physical port count per model
Throughput	CHR license limited	Line-rate physical forwarding
Management	GNS3 console	Dedicated management network
Monitoring	Wireshark ad-hoc	SNMP, Netflow, syslog
High Availability	Simulated	Redundant power, physical diversity

The configuration validated in this GNS3 lab transfers directly to physical MikroTik CCR hardware with no syntax changes, as CHR runs identical RouterOS. This makes the lab an effective pre-production validation environment.

9. Conclusion

This lab successfully implemented a 2x2 Clos network topology using MikroTik CHR in GNS3, achieving the primary objective of sub-second network convergence. Key accomplishments include:

- Designed and implemented a full 2x2 Clos fabric with proper spine-leaf architecture
- Achieved ~109ms failover time using OSPF + BFD — well within sub-second threshold
- Demonstrated ECMP load balancing across dual spine paths
- Implemented DHCP relay and DNS services across a routed fabric
- Validated convergence using Wireshark packet capture and continuous ping tests
- Identified and resolved practical challenges around BFD availability and OSPF network types

The lab proves that enterprise-grade network convergence is achievable using cost-effective MikroTik hardware, making Clos topology accessible beyond hyperscale data centers. The configuration is directly transferable to physical hardware, making this a practical pre-production validation exercise.

Key Takeaway

Sub-second network convergence is not exclusive to expensive Cisco Nexus or Arista hardware. With proper protocol configuration (OSPF + BFD) and appropriate hardware selection (MikroTik CHR/CCR), the same architectural benefits can be achieved at a fraction of the cost — making Clos topology practical for small and medium deployments.